Supplementary Materials for

Communicating Emotion Through Drawings Across Cultures

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S1. Study Materials and Coding Procedures

Table S1

Emotion Terms, Definitions, and Example Situations

Emotion	Definition	Example
Anger 愤怒	A feeling of displeasure resulting from injury, mistreatment, opposition, and usually showing itself in a desire to fight back at the supposed cause of this feeling. 一种因伤害、虐待、反对而产生的不愉快的感觉，通常表现为对认为造成这种感觉的原因进行反击的欲望。	The person received insulting words from others. 一个人收到了来自他人的侮辱性话语。
Disgust 厌恶	A sickening distaste, or dislike. 一种令人作呕的讨厌或不喜欢。	The person steps in dog feces, reaches down to wipe it off, and feces get on the person's hand. 一个人踩到了狗的粪便，伸手去擦，结果粪便沾到了这个人的手上。
Fear 恐惧	A feeling of anxiety and agitation caused by the presence or nearness of danger, evil, or pain. 由于危险、邪恶、痛苦的存在或接近而引起的焦虑和烦躁的感觉。	The person has realized that the brakes don't work while driving down a steep hill. The car approaches the end of the road, which is a cliff with no barrier. The person tries to brake and veers out of control. 一个人在把车开下陡峭的山坡时，已经意识到刹车不起作用。车子接近路的尽头，那是一个没有障碍物的悬崖。这个人试图刹车，但却失去了控制。
Happiness 快乐	Having a feeling of great pleasure, contentment, joy. 有一种非常高兴、满足、喜悦的感觉。	The person sees many close friends at a party. 一个人在一个聚会上看到了许多亲密的朋友。
Sadness 悲伤	Having low spirits or sorrow. 有低落的情绪或悲哀的感觉。	The person remembers the time last year when a young child died of a terminal illness. 一个人想起去年的时候，一个年轻的孩子死于绝症。
Surprise 惊讶	To come upon suddenly, or unexpectedly. 突然出现，或出乎意料地出现。	The person is looking at something new and unexpected. 一个人正看着新的、意想不到的东西。
Neutral 中性	Calm and no emotion was felt. 平静，没有感觉到任何情绪。	A person talked about the routine of his normal day. 一个人谈起他普通的一天的日程。

Note. These emotion terms, definitions, and examples were extracted from the Display Rule Inventory (Ekman & Cordaro, 2011; Matsumoto et al., 2005).

Table S2

Sixfold Self-Construal Scale

No.	English Version	Mandarin Version
	Personal Self–Construal	
1	I am a unique person separate from others.	我是一个与众不同的人。
2	I control my behavior to accommodate my wishes (interests).	我能控制自己的行为来满足自己的愿望。
3	I am affected by events that concern me.	我会受到与我有关的事情的影响。
4	I am aware of my own needs, desires and goals.	我了解自己的需求、欲望和目标。
5	I act as an independent person.	我作为一个独立的个体行事。
	Relational Self–Construal	
6	I think of myself as connected to my family.	我认为自己与周围关系亲密的人连接在一起。
7	I control my behavior to accommodate the wishes (interests) of my family.	我能够控制自己的行为以满足和我关系亲密的人的愿望。
8	I am affected by events that concern my family.	我会受到与我有亲密关系的人的事情的影响。
9	I am aware of the needs, desires and goals of my family.	我了解与我关系亲密的人的需求、要求和目标。
10	I feel I have a strong relationship with my family.	我认为自己与周围关系亲密的人之间存在牢固的联系。
	Collective Self–Construal	
11	I think of myself as connected to Canadian people.	我认为自己与中国人连接在一起。
12	I control my behavior to accommodate the wishes (interests) of Canadian people.	我能够控制自己的行为以满足中国人的愿望（需要）。
13	I am affected by events that concern Canadian people.	我会受到与中国人有关的事情的影响。
14	I am aware of the needs, desires and goals of Canadian people.	我了解中国人的需求、要求和目标。
15	I feel I have a strong relationship with Canadian people.	我认为自己与中国人之间存在牢固的联系。

Note. Responses were made on a 7-point scale ranging from 1 (*To a very small extent*) to 7 (*To a very large extent*). Items were drawn from the Sixfold Self–Construal Scale (Harb & Smith, 2008).

Table S3

Self-Construal Scale

No.	**English Version**	**Mandarin Version**
	Interdependent items	
1	I have respect for the authority figures with whom I interact.	我对跟我打交道的权威人士怀有敬意。
2	It is important for me to maintain harmony within my group.	对我来说，与他人维持融洽的关系非常重要。
3	My happiness depends on the happiness of those around me.	我的快乐取决于我周围人的快乐。
4	I would offer my seat in a bus to my professor.	乘车时我会主动为老师让座。
5	I respect people who are modest about themselves.	我尊重自身谦逊的人。
6	I will sacrifice my self–interest for the benefit of the group I am in.	为了集体的利益，我会牺牲自己的利益。
7	I often have the feeling that my relationships with others are more important than my own accomplishments.	我经常感到保持良好的人际关系比我自己取得的成绩更重要。
8	I should take into consideration my parents' advice when making education/career plans.	在制定教育/职业规划时，我应该考虑我父母的建议。
9	It is important to me to respect decisions made by the group.	对我来说尊重集体的决定是重要的。
10	I will stay in a group if they need me, even when I'm not happy with the group.	如果我所在的群体需要我，即使我待得不开心，我也仍然会留在那里。
11	If my brother or sister fails, I feel responsible.	如果我的兄弟姐妹失败，我会觉得负有责任。
12	Even when I strongly disagree with group members, I avoid an argument.	即使我非常不认同群体中其他成员的观点，我也会避免争论。
	Independent items	
13	I'd rather say “No" directly, than risk being misunderstood.	如果要说“不”，我宁愿直截了当而非隐晦含蓄，以免被误解。
14	Speaking up during a class is not a problem for me.	在课堂上大声发言对我来说不成问题。
15	Having a lively imagination is important to me.	对我来说，保持活跃的想象很重要。
16	I am comfortable with being singled out for praise or rewards.	当我被单独表扬或奖励时，我感到很自在。
17	I am the same person at home that I am at school.	我在家里和学校里的表现一致。
18	Being able to take care of myself is a primary concern for me.	对我来说，我首要关心的是能照顾好我自己。
19	I act the same way no matter who I am with.	不管和谁在一起我的表现始终如一。
20	I feel comfortable using someone's first name soon after I meet them, even when they are much older than I am.	在与某人认识不久后，我就能自然的直呼其名，即使对方比我年长很多。

No.	English Version	Mandarin Version
21	I prefer to be direct and forthright when dealing with people I've just met.	在与刚认识的人打交道时，我更喜欢直截了当。
22	I enjoy being unique and different from others in many respects.	我乐于在很多方面与众不同。
23	My personal identity independent of others, is very important to me.	独立于他人的个性特点对我来说是非常重要的。
24	I value being in good health above everything.	我认为健康比任何事情都重要。

Note. Responses were made on a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). Items were drawn from the Self–Construal Scale (Singelis, 1994)

Table S4

Analytic–Holistic Thinking Scale

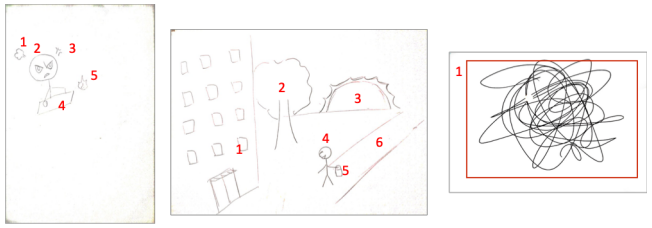
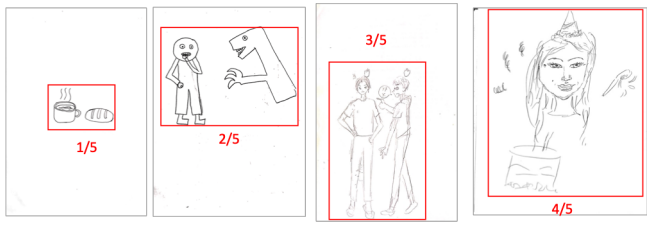
No.	English Version	Mandarin Version
	Causality	
1	Everything in the universe is somehow related to each other.	宇宙万物息息相关。
2	Nothing is unrelated.	没有任何事物是毫无关联的。
3	Everything in the world is intertwined in a causal relationship.	世间万物都在因果关系中相互交织。
4	Even a small change in any element of the universe can lead to significant alterations in other elements.	即使宇宙中任何一个元素的微小变化，都可能导致其他元素的重大改变。
5	Any phenomenon has numerous numbers of causes, although some of the causes are not known.	任何现象都有众多的成因，尽管有些成因尚不为人知。
6	Any phenomenon entails a numerous number of consequences, although some of them may not be known.	任何现象都会带来众多的后果，尽管其中一些可能不为人知。
	Attitude Toward Contradictions	
7	It is more desirable to take the middle ground than go to extremes.	采取中间立场比走极端更可取。
8	When disagreement exists among people, they should search for ways to compromise and embrace everyone's opinions.	当人们之间发生分歧时，应该寻求折衷的办法，并考虑每个人的意见。
9	It is more important to find a point of compromise than to debate who is right/wrong, when one's opinions conflict with other's opinions.	当个人观点与他人观点发生冲突时，找到折衷方案比争论对错更重要。
10	It is desirable to be in harmony, rather than in discord, with others of different opinions than one's own.	当自己和他人的观点不同时，保持和谐比据理力争更可取。
11	Choosing a middle ground in an argument should be avoided. *	在争论中应避免选择中间立场。
12	We should avoid going to extremes.	我们应该避免走极端。
	Perception of Change	
13	Every phenomenon in the world moves in predictable directions. *	世间万象都在朝着可预测的方向发展。
14	A person who is currently living a successful life will continue to stay successful. *	一个当下生活处于成功状态的人会一直成功下去。
15	An individual who is currently honest will stay honest in the future. *	一个当下诚实的人未来仍会保持诚实。
16	If an event is moving toward a certain direction, it will continue to move toward that direction. *	如果某事正朝着某个方向发展，它将继续朝着该方向发展下去。
17	Current situations can change at any time.	当前状况随时可能改变。
18	Future events are predictable based on present situations. *	未来的事件基于当下的形势是可预测的。
	Locus of Attention	

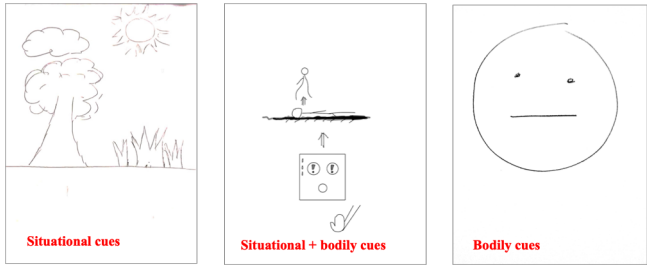
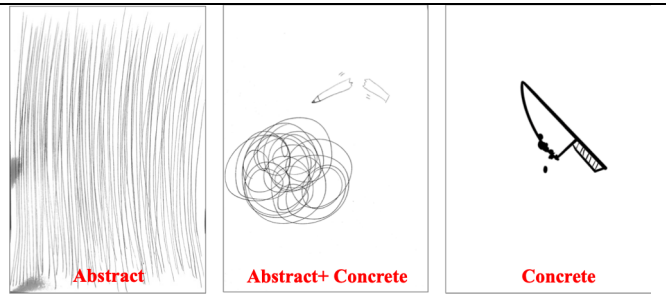
No.	English Version	Mandarin Version
19	The whole, rather than its parts, should be considered in order to understand a phenomenon.	要理解一种现象，应当考虑整体而非其部分。
20	It is more important to pay attention to the whole than its parts.	关注整体比关注其部分更为重要。
21	The whole is greater than the sum of its parts.	整体大于部分之和。
22	It is more important to pay attention to the whole context rather than the details.	关注整体情境而非细节更为重要。
23	It is not possible to understand the parts without considering the whole picture.	不考虑整体情况就无法理解各个部分。
24	We should consider the situation a person is faced with, as well as his/her personality, in order to understand one's behavior.	要理解一个人的行为，我们应该既考虑这个人所面对的情形，也考虑他/她的个性。

Note. Responses were made on a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). Items 11, 13–16, and 18 were reverse-coded prior to scoring. Items were drawn from the Analytic–Holistic Thinking Scale (Choi et al., 2007).

Table S5

Coding Manual for Primary Emotional Drawing Features

Features	Coding Guidelines	Examples
Object Count	1. A nameable and distinct region is coded as one object. 2. A complete object counts as a single unit regardless of internal features. *Examples*: • A house (with windows, door, roof) = 1 object • A tree (with trunk and leaves) = 1 object • A person (with face and body) = 1 object 3. Accessories that are part of the body (e.g., clothes, mask) are not counted separately unless detached (e.g., a bag or mask held in hand). 4. Effect symbols (e.g., question marks, exclamation marks, motion lines) are each coded as one object per type. 5. Abstract marks or visual elements without representational meaning (e.g., spirals) are counted as one object if they form a distinct unit.	
Page Coverage	1. Page coverage reflects the proportion of the paper occupied by visual content, coded in fifths (1/5 to 4/5). Estimation guide: a. 1/5 = a small portion of the page used b. 2/5 = slightly less than half the page c. 3/5 = slightly more than half the page d. 4/5 = most of the page used 2. If elements are scattered, estimate area as if the elements were grouped together.	

Features	Coding Guidelines	Examples
Bodily-versus-situational cues	Determine how emotion is conveyed: 1. 0 = Situational cues: Uses contextual cues or referential objects (e.g., storm clouds, knives) only. 2. 1 = Combines both situational and bodily cues. 3. 2 = Facial/bodily cues: Uses facial expressions or posture only.	
Abstraction-versus-concreteness	Evaluate the overall visual style: 1. 0 = Abstract: Non–representational lines/shapes 2. 1 = Mixed: Both abstract and concrete elements 3. 2 = Concrete: Clearly identifiable objects or people *Note*: This is based purely on visual form, not expressive intent.	

S2. Supplementary Analyses for Studies 1a and 1b

Table S6

Emotion-Inclusive Models for Drawing Features in Studies 1a and 1b

Drawing Feature	Effect	*df*	χ^2	*p*
Object count	Expresser Culture (EC)	1	10.68	.001
	Emotion (E)	6	73.58	< .001
	Drawing Batch (DB)	1	13.77	< .001
	EC × E	6	21.12	.002
	EC × DB	1	0.02	.876
	E × DB	6	20.46	.002
	EC × E × DB	6	4.10	.662
Page coverage	EC	1	4.26	.039
	E	6	84.47	< .001
	DB	1	6.89	.009
	EC × E	6	17.81	.007
	EC × DB	1	0.13	.714
	E × DB	6	12.87	.045
	EC × E × DB	6	12.77	.047
Situational/Bodily Expression	EC	1	26.47	< .001
	E	6	60.27	< .001
	DB	1	0.37	.544
	EC × E	6	17.08	.009
	EC × DB	1	0.59	.443
	E × DB	6	4.37	.626
	EC × E × DB	6	7.49	.278
Abstraction/Concreteness	EC	1	0.33	.569
	E	6	6.87	.333
	DB	1	0.39	.530
	EC × E	6	5.17	.522
	EC × DB	1	0.40	.525
	E × DB	6	8.08	.232
	EC × E × DB	6	8.98	.175

Figure S1

Emotion-Specific Cultural Differences in Drawing Features

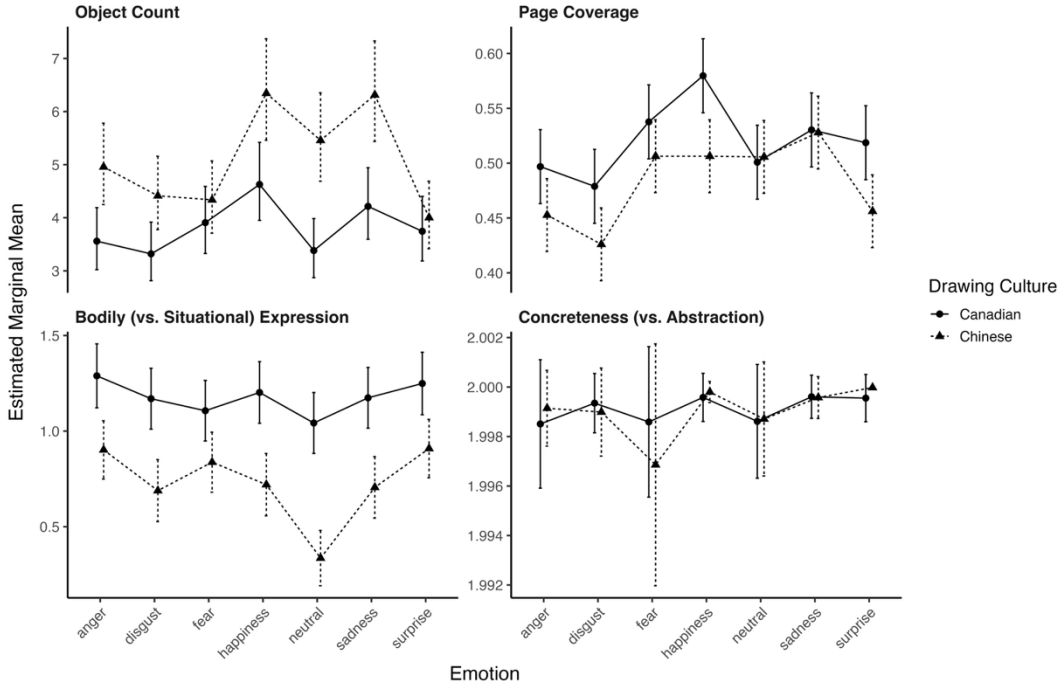


Table S7

Cultural Differences in Drawing Skill and Prior Drawing Training

Measures	Study	Canadian	Chinese	Statistic	*p*	Cohen's *d* [95% CI]
Drawing skill	1a	2.50 (1.70)	2.94 (1.51)	$t(205.22) = -1.97$	.050	–0.27 [–0.54, 0.00]
	1b	2.42 (1.35)	2.69 (1.43)	$t(85.99) = -0.91$	.364	–0.19 [–0.61, 0.23]
Drawing training	1a	16.19% (n = 17)	32.38% (n = 34)	$\chi^2(1) = 6.63$	.010	
	1b	13.95% (n = 6)	33.33% (n = 15)	$\chi^2(1) = 3.54$	.060	

Note. Values for drawing skill are means with standard deviations in parentheses. Values for drawing training are percentages of participants with prior training, with counts shown in parentheses.

Table S8

Descriptive Statistics and Group Comparisons for Cultural Measures

Study	Scale	Comparison	Group 1 *M (SD)*	Group 2 *M (SD)*	*t*(*df*)	*p*	Cohen's *d* [95% CI]
1a	SSCS Personal	Canadian vs. Chinese	5.54 (0.91)	5.38 (0.80)	1.32 (204.82)	.188	0.18 [–0.09, 0.45]
	SSCS Relational		5.56 (1.25)	5.31 (0.90)	1.67 (188.75)	.096	0.23 [–0.04, 0.50]
	SSCS Collective		4.60 (1.16)	5.49 (1.00)	–5.96 (203.21)	< .001	–0.82 [–1.10, –0.54]
1b	SCS	Canadian vs. Chinese	–0.02 (1.11)	0.19 (0.93)	–0.95 (82.02)	.343	–0.20 [–0.62, 0.22]
	AHS		4.84 (0.43)	5.32 (0.43)	–5.15 (85.77)	< .001	–1.10 [–1.54, –0.65]
2b & 3b	SCS	Canadian vs. Chinese	0.31 (1.08)	0.44 (0.86)	–0.66 (90.64)	.509	–0.14 [–0.54, 0.27]
		Canadian vs. Indian	0.31 (1.08)	0.16 (0.81)	0.75 (88.84)	.454	0.15 [–0.24, 0.55]
		Chinese vs. Indian	0.44 (0.86)	0.16 (0.81)	1.62 (91.99)	.108	0.33 [–0.07, 0.73]
	AHS	Canadian vs. Chinese	4.78 (0.50)	5.20 (0.43)	–4.42 (92.49)	< .001	–0.90 [–1.32, –0.48]
		Canadian vs. Indian	4.78 (0.50)	4.79 (0.49)	–0.15 (96.79)	.878	–0.03 [–0.43, 0.36]
		Chinese vs. Indian	5.20 (0.43)	4.79 (0.49)	4.35 (93.92)	< .001	0.88 [0.46, 1.30]

Note. SSCS = Sixfold Self-Construal Scale; SCS = Self-Construal Scale; AHS = Analytic–Holistic Thinking Scale. Values are means with standard deviations in parentheses. For each comparison, Group 1 and Group 2 correspond to the first and second cultural groups listed in the Comparison column, respectively. Group differences were tested using Welch's *t* tests.

Table S9

Configural Model Fit Indices for Cultural Measures

Study	Sample size	Scale	Configural χ^2 (*df*)	CFI	TLI	RMSEA[90% CI]	SRMR
1a	*N* = 210 (105 Canadian, 105 Chinese)	SSCS	381.72 (174)***	0.84	0.80	0.11 [0.09, 0.12]	0.10
1b	*N* = 88 (43 Canadian, 45 Chinese)	SCS	924.25 (502)***	0.33	0.27	0.14 [0.12, 0.15]	0.15
		AHS	916.75 (504)***	0.32	0.26	0.14 [0.12, 0.15]	0.14
2b & 3b	*N* = 145 (49 Canadian, 46 Chinese, 50 Indian)	SCS	1128.82 (753)***	0.56	0.51	0.10 [0.09, 0.11]	0.13
		AHS	1570.14 (756)***	0.32	0.25	0.15 [0.14, 0.16]	0.16

Note. SSCS = Sixfold Self-Construal Scale; SCS = Self-Construal Scale; AHS = Analytic-Holistic Thinking Scale; CFI = Comparative Fit Index; TLI = Tucker–Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual. Model fit was evaluated following Hu and Bentler (1999), with CFI/TLI ≥ .95 and RMSEA ≤ .06 indicating good fit. Across all scales and studies, configural models showed poor fit, so metric and scalar invariance were not tested. ***$p \leq .001$.

Table S10

Correlations Between Cultural Measures and Drawing Features

Study	Expresser Culture	Scale	Object Count	Page Coverage	Situational/ Bodily Expression	Abstraction/ Concreteness
1a	Canadian (N = 105)	SSCS Personal	–0.01	0.04	–0.09	–0.03
		SSCS Relational	0.03	–0.01	–0.12	0.08
		SSCS Collective	0.09	0.13	0.06	0.13
	Chinese (N = 105)	SSCS Personal	0.01	0.19	0.04	–0.02
		SSCS Relational	0.14	–0.02	0.10	–0.06
		SSCS Collective	0.07	–0.22	0.09	0.02
1b	Canadian (N = 43)	SCS	0.01	–0.18	–0.02	0.18
		AHS	0.05	–0.05	–0.06	0.17
	Chinese (N = 45)	SCS	–0.05	–0.07	–0.04	0.19
		AHS	0.32	0.31	–0.11	–0.08

Note. SSCS = Sixfold Self-Construal Scale; SCS = Self-Construal Scale; AHS = Analytic–Holistic Thinking Scale. Values are Pearson's correlation coefficients. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure separately within each Study × Expresser Culture family of tests. None of the FDR–adjusted p–values reached statistical significance, all ps ≥ .148.

S3. Supplementary Analyses for Studies 2a and 2b

Table S11

Emotion-Inclusive Models for Recognition Accuracy in Studies 2a and 2b

Effect	*df*	χ^2	*p*
Expresser Culture (EC)	1	0.91	.339
Perceiver Culture (PC)	1	2.73	.098
Emotion (E)	6	506.42	< .001
Drawing Batch (DB)	1	13.08	< .001
EC × PC	1	240.46	< .001
EC × E	6	13.32	.038
PC × E	6	226.42	< .001
EC × DB	1	2.76	.097
PC × DB	1	1.13	.287
E × DB	6	9.07	.169
EC × CP × E	6	97.07	< .001
EC × CP × DB	1	6.42	.011
EC × E × DB	6	10.99	.089
PC × E × DB	6	17.79	.007
EC × PC × E × DB	6	24.24	< .001

Figure S2

Emotion-Specific Recognition Patterns in Studies 2a and 2b

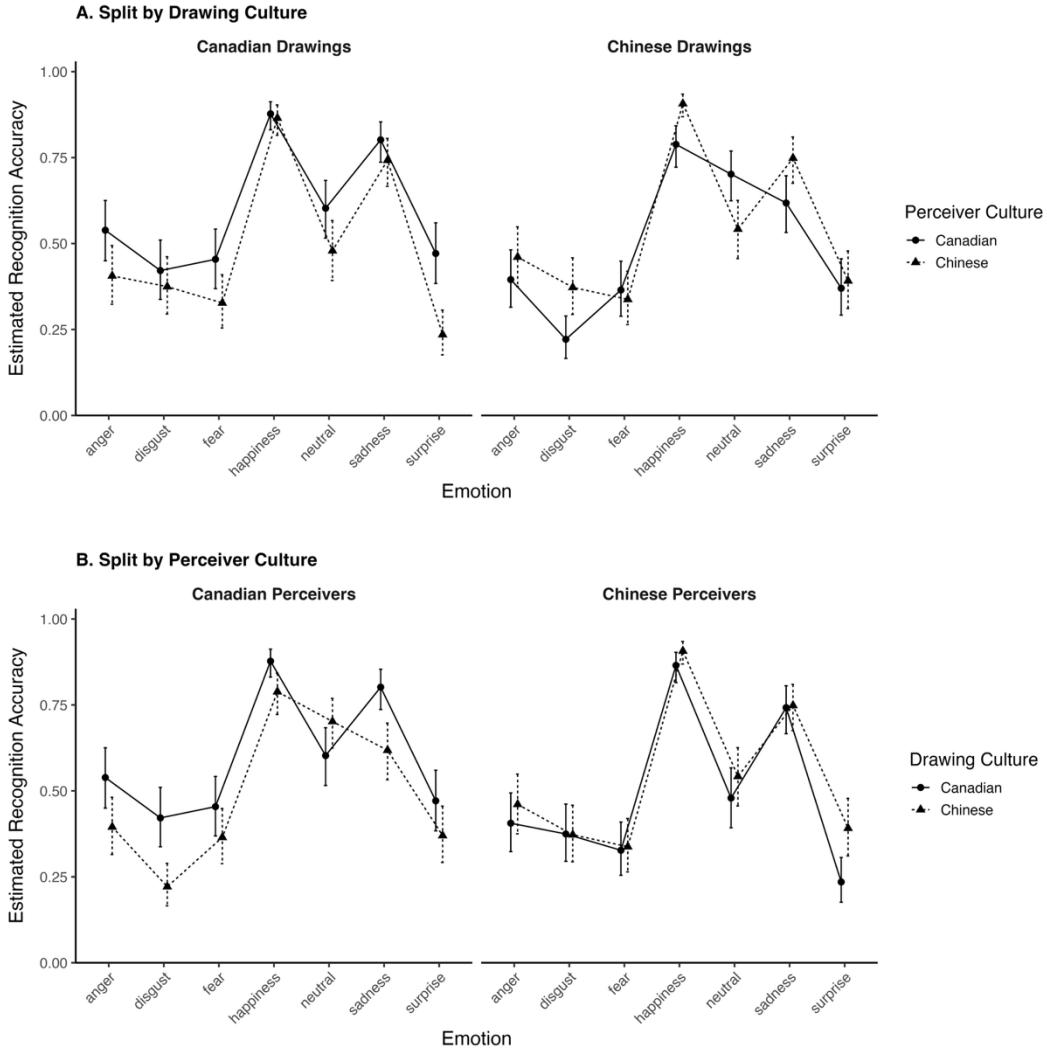


Table S12

Correlations Between Cultural Measures and Recognition Accuracy

Perceiver Culture	Scale	Overall Accuracy	Accuracy for Canadian Drawings	Accuracy for Chinese Drawings
Canadian (N = 49)	SCS	0.12	0.15	0.08
	AHS	0.02	0.02	0.02
Chinese (N = 46)	SCS	–0.02	–0.03	–0.01
	AHS	0.18	0.21	0.13
Indian (N = 50)	SCS	0.08	0.07	0.09
	AHS	–0.04	–0.04	–0.03

Note. SCS = Self-Construal Scale; AHS = Analytic–Holistic Thinking Scale. Values are Pearson's correlation coefficients. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure separately within each Perceiver Culture family of tests. None of the FDR–adjusted p–values reached statistical significance, all ps ≥ .693.

S4. Supplementary Analyses for Studies 3a and 3b

Table S13

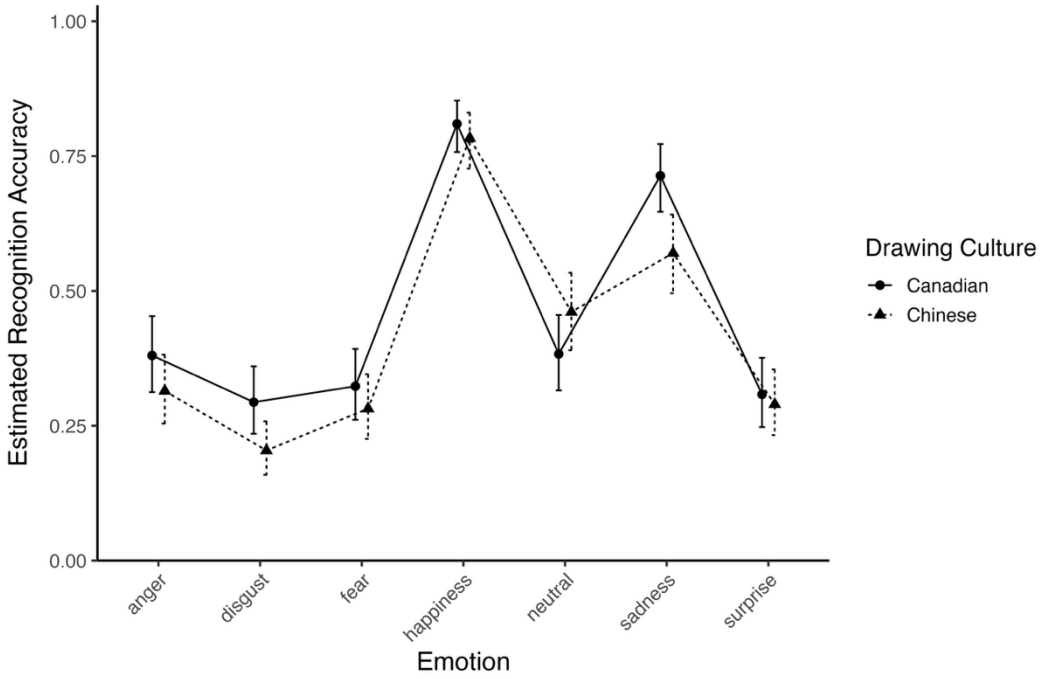
Emotion-Inclusive Models for Recognition Accuracy in Studies 3a and 3b

Effect	*df*	χ^2	*p*
Expresser Culture (EC)	1	4.05	.044
Emotion (E)	6	508.55	< .001
Drawing Batch (DB)	1	8.96	.003
EC × E	6	15.23	.019
EC × DB	1	3.66	.056
E × DB	6	5.25	.513
EC × E × DB	6	7.00	.321

Note. This model adds target emotion as a supplementary robustness check. The main effect of expresser culture did not reach significance in the primary model ($p = .052$) but reached significance after adjusting for emotion ($p = .044$). We therefore interpret the Indian-perceiver results as weak and model-dependent evidence for a possible overall Canadian-drawing advantage.

Figure S3

Emotion-Specific Recognition Patterns in Studies 3a and 3b



S5. Associations Between Primary Drawing Features and Recognition Outcomes

Table S14

Correlations Between Drawing Features and Recognition Accuracy Across Canadian, Chinese, and Indian Perceivers

Expresser Culture	Perceiver Group	Object Count	Page Coverage	Situational/ Bodily Expression	Abstraction/ Concreteness
Canadian	Canadian	< 0.01	0.04	0.19***	0.14***
	Chinese	–0.02	0.05	0.19***	0.12***
	Indian	0.04	0.05	0.17***	0.14***
Chinese	Canadian	–0.01	0.01	0.24***	0.10**
	Chinese	–0.01	–0.03	0.29***	0.11**
	Indian	0.01	–0.03	0.24***	0.10**

Note. Values are Spearman's rank-order correlations. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure separately within each Expresser Culture family of tests. Asterisks denote statistical significance based on FDR-adjusted p values: ** $p < .01$, *** $p < .001$.

Table S15

Correlations Between Drawing Features and Ingroup Advantage

Expresser Culture	Object Count	Page Coverage	Situational/ Bodily Expression	Abstraction /Concreteness
Canadian	0.05	–0.01	–0.03	< 0.01
Chinese	0.03	–0.06	0.08	0.02

Note. Values are Spearman's rank-order correlations. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure. None of the FDR–adjusted *p*–values reached statistical significance, all *p*s ≥ .058.

S6. Additional Coding of Subject–Background Relations

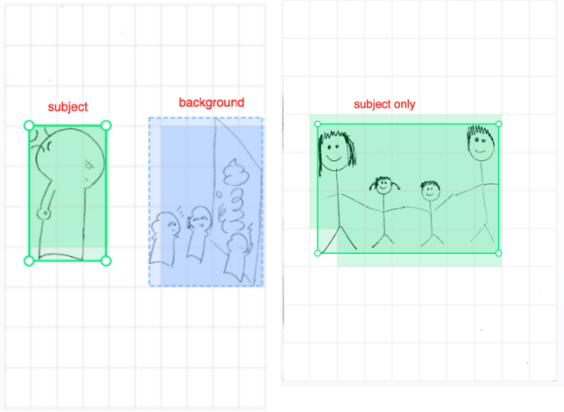
We conducted an additional exploratory coding of drawings that conveyed emotion through both bodily and situational cues (n = 878). This subset was selected because spatial comparisons between subject and background are meaningful only when both are present.

Two coders used a custom browser-based annotation tool to identify the subject region and the non-subject background region in each drawing. Coders manually enclosed each region using rectangular selections. The tool computed each selected rectangle's area relative to the total image area, yielding proportional area values ranging from 0 to 1. Subject size and background size were operationalized as these proportional area values. The subject-to-background ratio was computed as subject area divided by background area and was log-transformed before analysis to address right-skewed distributions.

Two coders independently coded the drawings. Ten percent of the full dataset was coded by both coders to assess reliability, with the remaining drawings split between the two coders. Intraclass correlation coefficients indicated good to excellent reliability for subject size, ICC = 0.77, 95% CI [0.67, 0.84], background size, ICC = 0.89, 95% CI [0.83, 0.92], and subject-to-background ratio, ICC = 0.78, 95% CI [0.69, 0.85].

Table S16

Coding Manual for Additional Subject–Background Features

Feature	Coding Guidelines	Examples
Subject Size	The subject is defined as the part of the drawing most directly expressing the target emotion. • If a figure is clearly expressing the target emotion, that figure is treated as the subject. • If multiple elements jointly express the target emotion, all are selected together as a single subject. • Speech bubbles, text, and symbols are excluded unless they directly constitute the emotional expression.	
Background Size	Background refers to all elements other than the subject • If the subject occupies the entire drawing with no other visible elements, the drawing is coded as having no visible background. • Speech bubbles, text, or symbols that do not directly carry the target emotion are included in the background.	
Subject-to-Background Ratio	The ratio of the area occupied by the subject to the area occupied by the background.	

We analyzed the two drawing batches together using linear mixed-effects models. Each model included expresser culture, drawing batch, and their interaction as fixed effects, with a random intercept for expresser ID to account for multiple drawings produced by each expresser. As shown in Table S17, expresser culture significantly predicted subject size, such that Canadian drawings had larger emotion-relevant subjects than Chinese drawings (Canadian: $M = 0.13$, $SE = 0.01$; Chinese: $M = 0.10$, $SE = 0.01$), $\chi^2(1) = 4.96$, $p = .026$, $b = 0.03$, 95% CI [0.00, 0.05]. Expresser culture did not significantly predict background size, $\chi^2(1) = 2.01$, $p = .156$, or subject-to-background ratio, $\chi^2(1) = 0.23$, $p = .630$. None of these effects were qualified by drawing batch.

Table S17

Effects of Expresser Culture, Drawing Batch, and Their Interaction on Additional Drawing Features in Studies 1a and 1b

Drawing Feature	Effect	*df*	χ^2	*p*
Subject Size	Expresser Culture (EC)	1	4.96	.026
	Drawing Batch (DB)	1	1.23	.266
	EC × DB	1	1.53	.216
Background Size	EC	1	2.01	.156
	DB	1	0.43	.512
	EC × DB	1	0.82	.365
Subject-to-Background	EC	1	0.23	.630
Ratio	DB	1	0.99	.319
	EC × DB	1	1.65	.199

Table S18

Correlations Between Additional Subject–Background Features and Recognition Accuracy Across Canadian, Chinese, and Indian Perceivers

Expresser Culture	Perceiver Group	Subject Area	Background Area	Subject-to-Background Ratio
Canadian	Canadian	0.20***	–0.06	0.18***
	Chinese	0.20***	–0.07	0.19***
	Indian	0.18***	–0.01	0.12*
Chinese	Canadian	0.26***	0.03	0.15**
	Chinese	0.23***	–0.01	0.15**
	Indian	0.22***	–0.02	0.15**

Note. Values are Spearman's rank-order correlations. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure separately within each Expresser Culture family of tests. Asterisks denote statistical significance based on FDR-adjusted p values: *$p < .05$,** $p < .01$, *** $p < .001$.

Table S19

Correlations Between Additional Subject–Background Features and Ingroup Advantage

Expresser Culture	Subject Area	Background Area	Subject-to-Background Ratio
Canadian	–0.11	0.02	0.01
Chinese	–0.02	–0.06	–0.06

Note. Values are Spearman's rank-order correlations. False discovery rate (FDR) correction was applied using the Benjamini–Hochberg procedure. None of the FDR–adjusted *p*–values reached statistical significance, all *p*s > .074.

S7. Interactions Between Drawing Features and Perceiver Cultural Orientation in Recognition Accuracy

We examined whether the associations between drawing features and trial-level recognition accuracy varied with perceivers' measured cultural orientations in Studies 2b and 3b, the only recognition studies in which the AHS and SCS were administered. Separate binomial generalized linear mixed-effects models were fitted for each combination of drawing feature and cultural-orientation measure. The four primary coded features—object count, page coverage, situational/bodily expression, and abstraction/concreteness—were each examined in relation to AHS and SCS scores. In Study 2b, AHS and SCS scores were standardized within perceiver culture; in Study 3b, they were standardized within the Indian perceiver sample.

For Study 2b, the models were specified as follows: Accuracy ~ Expresser Culture × Perceiver Culture + Drawing Feature × Perceiver Cultural Orientation + (1 | Perceiver ID) + (1 | Expresser ID) + (1 | Stimulus). For Study 3b, in which all perceivers were Indian, the models were specified as follows: Accuracy ~ Expresser Culture + Drawing Feature × Perceiver Cultural Orientation + (1 | Perceiver ID) + (1 | Expresser ID) + (1 | Stimulus). The Drawing Feature × Perceiver Cultural Orientation interaction tested whether the association between a given drawing feature and recognition accuracy varied with perceivers' cultural orientations. When the variance of the Expresser ID random intercept was estimated at approximately zero, that random effect was omitted from the model.

As shown in Table S20, none of the Drawing Feature × Perceiver Cultural Orientation interactions for the primary coded features were significant in either Study 2b or Study 3b (all *p*s > .05). Thus, the associations between the primary

drawing features and recognition accuracy did not reliably vary with AHS or SCS scores.

Table S20

Drawing Feature × Perceiver Cultural Orientation Interactions for Primary Drawing Features in Studies 2b and 3b

Study	Drawing Feature	Scale	*df*	χ^2	*p*
Study 2b	Object Count	AHS	1	3.45	.063
		SCS	1	0.04	.844
	Page Coverage	AHS	1	1.76	.184
		SCS	1	0.47	.491
	Situational/Bodily Expression	AHS	2	0.05	.974
		SCS	2	2.26	.323
	Abstraction/Concreteness	AHS	2	0.47	.792
		SCS	2	1.98	.372
Study 3b	Object Count	AHS	1	1.68	.194
		SCS	1	0.98	.322
	Page Coverage	AHS	1	0.91	.341
		SCS	1	0.49	.485
	Situational/Bodily Expression	AHS	2	1.06	.589
		SCS	2	3.82	.148
	Abstraction/Concreteness	AHS	2	1.04	.594
		SCS	2	0.91	.634

Note. AHS = Analytic–Holistic Thinking Scale; SCS = Self-Construal Scale. The χ^2 statistic tests the Drawing Feature × Perceiver Cultural Orientation interaction.

We conducted the same analyses for the additional subject–background features. As shown in Table S21, none of the Drawing Feature × Perceiver Cultural Orientation interactions was significant in either study (all *p*s > .05).

Table S21

Drawing Feature × Perceiver Cultural Orientation Interactions for Additional Subject–Background Features in Studies 2b and 3b

Study	Drawing Feature	Scale	*df*	χ^2	*p*
Study 2b	Subject Size	AHS	1	0.83	.363
		SCS	1	0.28	.598
	Background Size	AHS	1	0.80	.372
		SCS	1	0.63	.426
	Subject-to-Background Ratio	AHS	1	1.21	.272
		SCS	1	1.84	.174
Study 3b	Subject Size	AHS	1	0.50	.478
		SCS	1	0.19	.660
	Background Size	AHS	1	0.05	.821
		SCS	1	0.18	.674
	Subject-to-Background Ratio	AHS	1	0.21	.646
		SCS	1	0.57	.452

Note. AHS = Analytic–Holistic Thinking Scale; SCS = Self-Construal Scale. The χ^2 statistic tests the Drawing Feature × Perceiver Cultural Orientation interaction.

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